Differential Equation Descriptions:
Continuous-Time Systems with Stored Energy

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Key Concepts

1) Differential equations are one of the few descriptions that allow analysis of a system with stored energy at the time an input is applied.

2) Differential equation

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

   a) Input $x(t)$, output $y(t)$

b) Typically $N \geq M$ so the order of difference equation is N

c) Initial conditions

$$y(t)|_{t=0^-},\ \frac{d}{dt}y(t)\Big|_{t=0^-},\ \frac{d^2}{dt^2}y(t)\Big|_{t=0^-},\ \frac{d^{N-1}}{dt^{N-1}}y(t)\Big|_{t=0^-}$$

i) Summarize the entire history of the system prior to $t = 0$. Nothing more needs to be known about the past to determine current and future output.

ii) If they are zero, the system is at rest.

iii) If they are non zero, they represent the energy stored in the system due to past inputs.

3) Analytical approach for solving differential equations gives insight into the relationship between the coefficients $\{b_0, \ldots, b_M, a_1, \ldots, a_N\}$ and the system response. We assume the input is applied at time $t = 0$.

4) The difference equation is linear so the output can be written as a sum of two terms $y(t) = y_s(t) + y_t(t), \;\; t > 0$.

    a) Steady state response $y_s(t)$: output when the input is applied for all time.

    b) Transient response $y_t(t)$: how the system transitions from its initial condition at $t = 0$ to steady state.

5) Steady state response $y_s(t)$.

a) Assume the input is a sum of complex exponential signals

$$x(t) = \sum_{i=1}^{P} \alpha_i e^{s_i t}$$

where s_i is a complex number.

b) We only need the output for an input $x(t) = e^{st}$. The output is $H(s)e^{st}$ where

$$H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$$

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Then by linearity

$$y_s(t) = \sum_{i=1}^{P} \alpha_i H(s_i) e^{s_i t}$$

c) $H(s)$ is called the transfer function of the system.

6) Transient response $y_t(t)$ is the system output when the input is 0. This follows from linearity and the fact that $y_s(t)$ is the output for the specified input.

a) The transient response is

$$y_t(t) = \sum_{\ell=1}^{N} c_\ell e^{\tilde{s}_\ell t}$$

b) The $\tilde{s}_\ell$ are the N roots of the system's characteristic equation

$$\sum_{k=0}^{N} a_k s^k = 0$$

c) The constants c_ℓ are chosen so that $y(t) = y_s(t) + y_t(t)$ satisfies the initial conditions.

d) This form of the solution applies when the roots $\tilde{s}_\ell$ are distinct and not equal to any of the s_i in the input $x(t)$.

7) The solution $y(t)$ is only valid for $t > 0$.

a) If the right hand side of the differential equation, $\sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$, contains no impulses or derivatives of impulses, then the initial conditions at $t = 0^+$ are identical to those for $t = 0^-$.

b) If the right hand side of the differential equation, $\sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$, contains an impulse or derivative of an impulse, then the initial conditions at $t = 0^-$ must be translated forward to $t = 0^+$. This procedure is beyond the scope of the lecture.

8) Given $y_s(t)$ and $s_1, s_2, \ldots s_N$, we obtain c_ℓ by solving the system of

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N linear equations

$$y(0^+) = \sum_{i=1}^{P} \alpha_i H(s_i) + c_1 + c_2 + \ldots + c_N$$

$$\frac{d}{dt} y(t)\Big|_{t=0^+} = \sum_{i=1}^{P} \alpha_i s_i H(s_i) + c_1 \tilde{s}_1 + c_2 \tilde{s}_2 + \ldots + c_N \tilde{s}_N$$

$$\vdots$$

$$\frac{d^{N-1}}{dt^{N-1}} y(t)\Big|_{t=0^+} = \sum_{i=1}^{P} \alpha_i s_i^{N-1} H(s_i) + c_1 \tilde{s}_1^{N-1} + c_2 \tilde{s}_2^{N-1} + \ldots + c_N \tilde{s}_N^{N-1}$$

where $\tilde{s}_\ell$ are the roots of the characteristic equation.

Video Screenshots

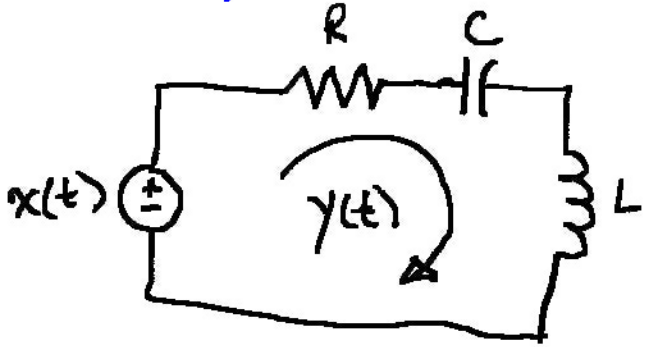
Linear, constant coefficient differential equations describe linear, time-invariant continuous-time systems

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

typically $N \geq M$.

Order is N - the number of energy storage devices

Example:



$$R y(t) + \frac{1}{C}\int_{-\infty}^{t} y(\tau)\,d\tau + L\frac{dy(t)}{dt} = x(t)$$

take $\frac{d}{dt}$

$$\frac{1}{C} y(t) + R\frac{d}{dt} y(t) + L\frac{d^2}{dt^2} y(t) = \frac{d}{dt} x(t)$$

$a_0 = \frac{1}{C}$, $a_1 = R$, $a_2 = L$, $b_0 = 0$, $b_1 = 1$

Initial conditions

$$y(t)\Big|_{t=0^-},\ \frac{d}{dt}y(t)\Big|_{t=0^-},\ \frac{d^2}{dt^2}y(t)\Big|_{t=0^-}\ \cdots\ \frac{d^{N-1}}{dt^{N-1}}y(t)\Big|_{t=0^-}$$

- directly related to initial values of energy storage devices
- summarize all information about the system's past needed to determine future outputs.

Solutions:

$$y(t) = y_s(t) + y_t(t),\quad t > 0$$

1) Steady-state response - $y_s(t)$
   as if input has been applied forever

2) Transient response - $y_t(t)$
   determines how system transitions from initial state to steady-state response

Steady-State Response

Key input: $x(t) = e^{st}$ (s a complex number)

$\Rightarrow y_s(t) = H(s)\, e^{st}$

complex exponential input $\Rightarrow$ complex exponential output

Check: $\frac{d^k}{dt^k} x(t) = s^k e^{st}$; $\frac{d^k}{dt^k} y_s(t) = s^k H(s) e^{st}$

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y_s(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t) \Rightarrow e^{st} H(s) \sum_{k=0}^{N} a_k s^k = e^{st} \sum_{k=0}^{M} b_k s^k$$

Thus, $H(s) = \dfrac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$ system transfer function

By linearity: $x(t) = \sum_{q=1}^{Q} \alpha_q e^{s_q t} \longrightarrow y_s(t) = \sum_{q=1}^{Q} \alpha_q H(s_q) e^{s_q t}$

Sum of exponentials $\longrightarrow$ sum of exponentials

Transient Response

$y(t) = y_s(t) + y_t(t)$

If $y_s(t)$ satifies the diff. equation for input $x(t)$, by linearity $y_t(t)$ satisfies the diff. equation for input $x(t) = 0$

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y_t(t) = 0$$

$$y_t(t) = \sum_{\ell=1}^{N} c_\ell e^{s_\ell t}$$ where s_ℓ are roots of $a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0 = 0$

"characteristic equation"

(assuming non repeated roots)

Check: $\frac{d^k}{dt^k} e^{s_\ell t} = s_\ell^k e^{s_\ell t}$ $\Rightarrow$ $\sum_{k=0}^{N} a_k \frac{d^k}{dt^k}(e^{s_\ell t}) = e^{s_\ell t} \underbrace{\sum_{k=0}^{N} a_k s_\ell^k}_{=0}$

Constants c_ℓ are chosen so $y(t)$ satisfies the initial conditions

Putting it all together-

$x(t) = \sum_{q=1}^{Q} \alpha_q e^{s_q t}$ (α_q, s_q known)

$$y(t) = \sum_{q=1}^{Q} \alpha_q H(s_q) e^{s_q t} + \sum_{\ell=1}^{N} c_\ell e^{s_\ell t}, \quad t > 0$$

where $H(s) = \dfrac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$, $s_\ell, \ell = 1, 2, \ldots N$ satisfy $a_N s^N + \ldots + a_1 s + a_0 = 0$

Note $\left.\dfrac{d^k}{dt^k} y(t)\right|_{t=0} = \sum_{q=1}^{Q} \alpha_q H(s_q) s_q^k + \sum_{\ell=1}^{N} c_\ell s_\ell^k$

- Use N initial conditions to get N equations for the c_ℓ and solve
- If $\sum_{k=0}^{M} b_k \dfrac{d^k}{dt^k} x(t)$ contains impulses or derivatives, initial conditions at $t = 0^-$ must be translated to $t = 0^+$ before solving for c_ℓ